VIRTUALWORKS LAB INTERNSHIP

PHARMACOVIGILANCE

TASK 4

SEVERITY & CASE CLASSIFICATION

Classification of ADRs by Seriousness, Expectedness and Clinical Impact

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Note: All patient cases in this report are fictional and are prepared only for internship learning and pharmacovigilance classification practice. Expectedness is assessed against the academic reference safety information stated in each case, not against any specific commercial product label.

1. Aim and Objective

Aim: To classify adverse drug reactions according to seriousness, expectedness and clinical impact using standard pharmacovigilance concepts.

Objective: To review multiple patient case scenarios, identify whether each reaction is serious or non-serious, decide whether it is expected or unexpected against the stated reference safety information, and describe the practical severity and clinical impact of the event.

2. Key Concepts Used for Classification

2.1 Seriousness

In pharmacovigilance, seriousness is a regulatory classification based on the outcome or consequence of an adverse event. A reaction is considered serious when it results in death, is life-threatening, requires inpatient hospitalization or prolongs hospitalization, causes persistent or significant disability/incapacity, causes a congenital anomaly/birth defect, or is another medically important event that may require intervention to prevent one of these outcomes.

2.2 Severity

Severity describes the intensity of the reaction and is not the same as seriousness. For this educational task, I used a simple practical scale: mild = minimal effect on daily activity; moderate = noticeable symptoms requiring treatment or a change in therapy; severe = marked symptoms, major functional limitation, emergency care or hospital-level management. A severe reaction is not automatically serious, and a serious event may not always be described as severe at every stage.

2.3 Expectedness

Expectedness asks whether the nature and clinical pattern of the reaction are consistent with the applicable reference safety information. For these fictional cases, each scenario states whether the event is included in the academic reference safety information. If it is listed, I classify it as expected; if it is not listed or differs meaningfully from the stated information, I classify it as unexpected.

3. Practical Classification Guide

Classification	How I used it
Serious	At least one regulatory seriousness criterion is met.
Non-serious	No regulatory seriousness criterion is met.
Expected	Reaction is consistent with the academic reference safety information in the case.
Unexpected	Reaction is not described in the academic reference safety information in the case.
Mild clinical impact	Little disruption; observation or minimal treatment only.
Moderate clinical impact	Requires symptomatic treatment, monitoring or treatment adjustment.
High clinical impact	Emergency intervention, hospitalization or major risk to the patient.

4. Case Studies and Classification

1. Cetirizine-Associated Drowsiness

A 22-year-old woman started cetirizine 10 mg once daily for allergic rhinitis. A few hours after the first dose she felt sleepy and less alert. She remained able to perform normal activities and did not require medical treatment. The symptoms reduced after she avoided taking the medicine during daytime. The academic reference safety information for this case lists drowsiness/somnolence as a known reaction.

Assessment point	Finding
Reaction	Drowsiness / somnolence
Severity	Mild
Seriousness	Non-serious - no seriousness criterion met
Expectedness	Expected - listed in the case reference information
Clinical impact	Low; temporary effect on alertness
Action	Timing of dose adjusted; monitoring advised

My assessment: This is a mild, non-serious and expected ADR. The main clinical concern is reduced alertness, so counseling about driving or operating machinery is important even though the event does not meet serious criteria.

2. Amoxicillin-Clavulanate-Associated Diarrhea

A 31-year-old man received amoxicillin + clavulanate for a dental infection. On day 3 he developed four loose stools in 24 hours with mild abdominal discomfort. He remained hydrated, had no blood in stool, no fever and no hospital admission. Oral fluids and dietary advice were given. The symptoms improved after completion of therapy. The academic reference information lists diarrhea and gastrointestinal upset as known reactions.

Assessment point	Finding
Reaction	Diarrhea with mild abdominal discomfort
Severity	Moderate
Seriousness	Non-serious
Expectedness	Expected
Clinical impact	Moderate; required advice and monitoring but no admission
Action	Hydration, dietary advice and observation

My assessment: The event affected comfort and required management, so I considered the intensity moderate. However, it did not cause hospitalization, life-threatening illness or another serious outcome. Because diarrhea is included in the academic reference information, the event is expected.

3. Insulin-Associated Severe Hypoglycemia

A 58-year-old man using short-acting insulin accidentally took his usual dose but skipped his meal. He became confused, lost consciousness and was taken to the emergency department. His capillary blood glucose was 34 mg/dL. Intravenous glucose was given and he was admitted overnight for observation. He recovered fully. The academic reference information lists hypoglycemia as a known risk of insulin.

Assessment point	Finding
Reaction	Severe hypoglycemia with loss of consciousness
Severity	Severe
Seriousness	Serious - life-threatening event and hospitalization
Expectedness	Expected
Clinical impact	High; emergency treatment and admission required
Action	IV glucose, observation and insulin-use counseling

My assessment: This case is both severe and serious. Loss of consciousness and hospital admission clearly meet seriousness criteria. The reaction is expected because hypoglycemia is included in the stated reference safety information, but expected events can still be clinically serious and require urgent reporting and management.

4. Carbamazepine-Associated Stevens-Johnson Syndrome

A 40-year-old woman started carbamazepine for trigeminal neuralgia. Twelve days later she developed fever, painful widespread rash and oral mucosal erosions. She was admitted to hospital and carbamazepine was immediately withdrawn. Dermatology review supported Stevens-Johnson syndrome. She required supportive inpatient treatment. In this academic case, Stevens-Johnson syndrome is included in the reference safety information as a rare but important known reaction.

Assessment point	Finding
Reaction	Stevens-Johnson syndrome
Severity	Severe
Seriousness	Serious - hospitalization and medically important event
Expectedness	Expected
Clinical impact	High; potential for major morbidity and progression
Action	Drug withdrawn; urgent inpatient supportive care

My assessment: This is a serious and severe ADR because it required hospitalization and is medically important. It remains expected in this academic case because the reaction is specifically included in the stated reference information. Expectedness does not reduce the seriousness of the event.

5. Atorvastatin and Acute Pancreatitis - Unexpected Academic Case

A 52-year-old man taking atorvastatin for dyslipidemia developed sudden severe upper abdominal pain and repeated vomiting three weeks after treatment was started. Serum lipase was markedly elevated and imaging supported acute pancreatitis. He was admitted for intravenous fluids, analgesia and monitoring. No gallstones were seen and he denied alcohol use. For this academic scenario only, the provided reference safety information lists myalgia, headache and mild liver-enzyme elevation but does not list pancreatitis.

Assessment point	Finding
Reaction	Acute pancreatitis
Severity	Severe
Seriousness	Serious - inpatient hospitalization
Expectedness	Unexpected in the provided academic reference information
Clinical impact	High; required diagnostic work-up and inpatient management
Action	Suspected drug withheld; supportive treatment

My assessment: The hospitalization makes this a serious case. Because pancreatitis is not included in the academic reference safety information supplied for this scenario, I classify it as unexpected. In real pharmacovigilance work, expectedness must be checked against the current approved reference information rather than assumed from memory.

6. Omeprazole-Associated Generalized Urticaria - Unexpected Academic Case

A 27-year-old woman started omeprazole for dyspepsia. Within two days she developed widespread itchy wheals over the trunk and arms. She had no breathing difficulty, facial swelling, hypotension or mucosal involvement. Omeprazole was stopped and an oral antihistamine was prescribed. The rash resolved over the next 24 hours without emergency admission. For this academic case, generalized urticaria is not included in the provided reference safety information.

Assessment point	Finding
Reaction	Generalized urticaria
Severity	Moderate
Seriousness	Non-serious - no serious outcome occurred
Expectedness	Unexpected in the provided academic reference information
Clinical impact	Moderate; required drug withdrawal and symptomatic treatment
Action	Drug stopped; oral antihistamine given

My assessment: This reaction required treatment and a change in medication, so its clinical impact is moderate. It did not meet regulatory seriousness criteria. Under the reference information given for this academic exercise, it is unexpected.

5. Consolidated Case Classification

Case	ADR	Severity	Seriousness	Expectedness	Clinical impact
1	Drowsiness	Mild	Non-serious	Expected	Low
2	Diarrhea	Moderate	Non-serious	Expected	Moderate
3	Severe hypoglycemia	Severe	Serious	Expected	High
4	Stevens-Johnson syndrome	Severe	Serious	Expected	High
5	Acute pancreatitis	Severe	Serious	Unexpected*	High
6	Generalized urticaria	Moderate	Non-serious	Unexpected*	Moderate

**Expectedness in Cases 5 and 6 is based only on the fictional academic reference safety information provided within those cases.*

6. What I Learned From the Cases

The cases show that seriousness, severity and expectedness answer different questions. A reaction can be expected but still serious, as seen with severe hypoglycemia and Stevens-Johnson syndrome. A reaction can also be unexpected without being serious, as in the academic urticaria example. For correct case classification, I would first identify the clinical event, then check the regulatory seriousness criteria, judge intensity separately, and finally compare the reaction with the applicable reference safety information.

7. Conclusion

In this task, six ADR case scenarios were reviewed and classified by seriousness, severity, expectedness and clinical impact. Seriousness was determined from patient outcomes and regulatory criteria, while severity was used to describe the intensity of each reaction. Expectedness was decided by comparing each event with the academic reference safety information supplied in the case. This approach helps prioritize cases that may require urgent medical attention, expedited reporting or further safety review.

8. References

1. International Council for Harmonisation (ICH). E2A: Clinical Safety Data Management - Definitions and Standards for Expedited Reporting.

2. Pharmacovigilance Programme of India (PvPI), Indian Pharmacopoeia Commission. Pharmacovigilance Toolkit - definitions of serious and unexpected adverse reactions.

3. Pharmacovigilance Programme of India (PvPI). Frequently Asked Questions - serious ADR / serious adverse event criteria.

4. ICH E6(R3) Guideline. Glossary: adverse event, adverse drug reaction, serious adverse event and SUSAR definitions.

Academic integrity note: The patient scenarios are fictional and are used only to demonstrate pharmacovigilance classification. No real patient identifiers or confidential health information are included.